

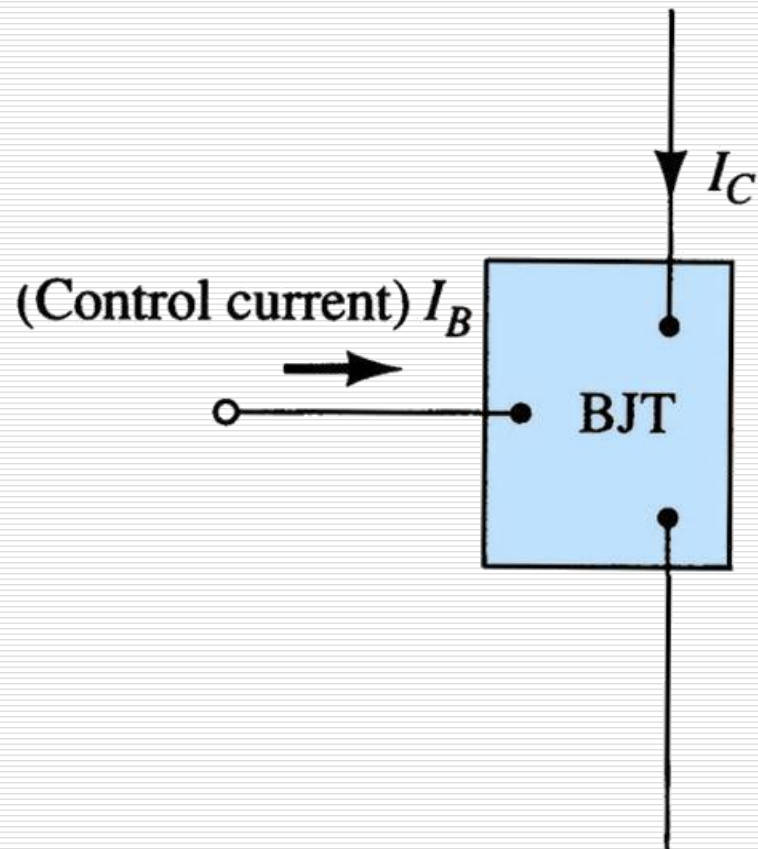
JFET

Junction Field Effect Transistor

Introduction (FET)

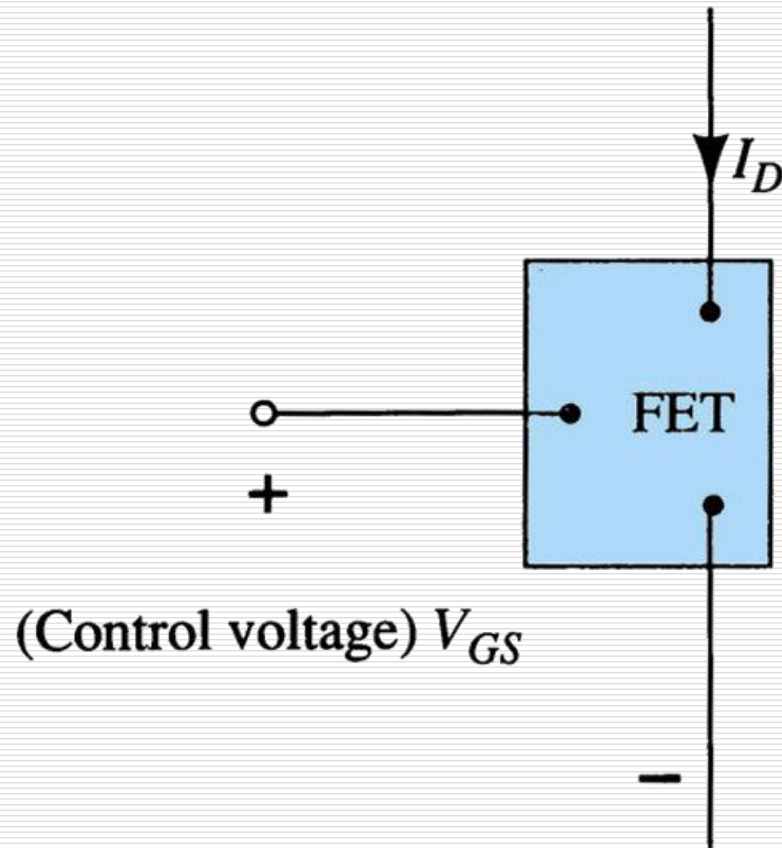
- ❑ Field-effect transistor (FET) are important devices such as BJTs
 - ❑ Also used as amplifier and logic switches
 - ❑ What is the difference between JFET and BJT?
-

BJT is Current-controlled



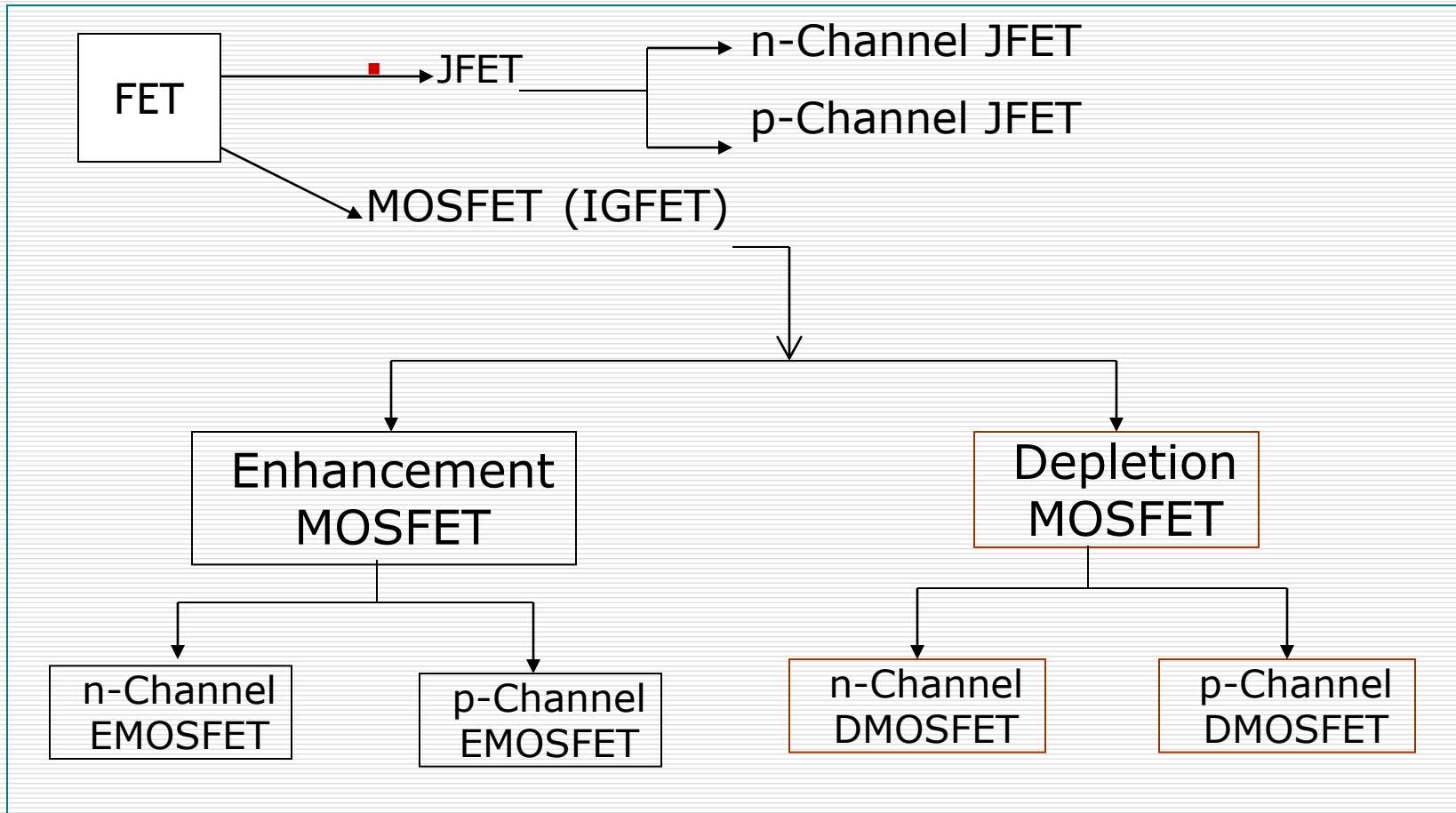
(a)

FET is Voltage-controlled



(b)

Types of Field Effect Transistors (The Classification)



Introduction.. (Advantages of FET over BJT)

- ❑ High input impedance ($M\Omega$)
(Linear AC amplifier system)
- ❑ Temperature stable than BJT
- ❑ Smaller than BJT
- ❑ Can be fabricated with fewer processing
- ❑ BJT is bipolar – conduction both hole and electron
- ❑ FET is unipolar – uses only one type of current carrier
- ❑ Less noise compare to BJT
- ❑ Usually use as an Amplifier and logic switch

Disadvantages of FET

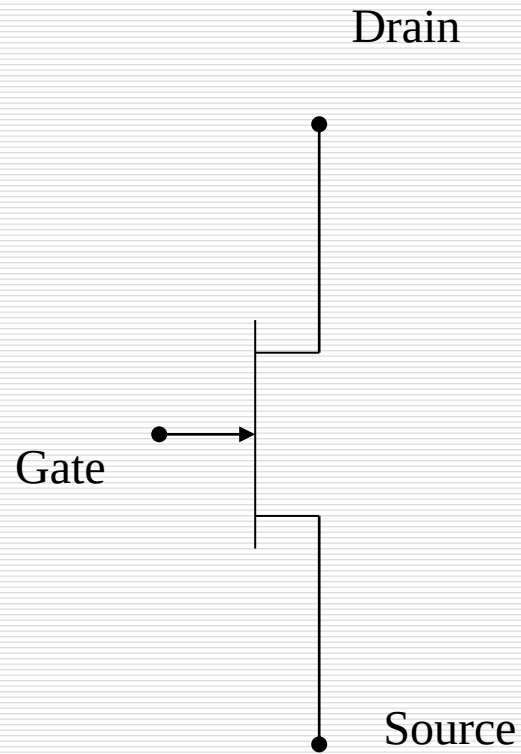
- ❑ Easy to damage compare to BJT

Junction field-effect transistor..

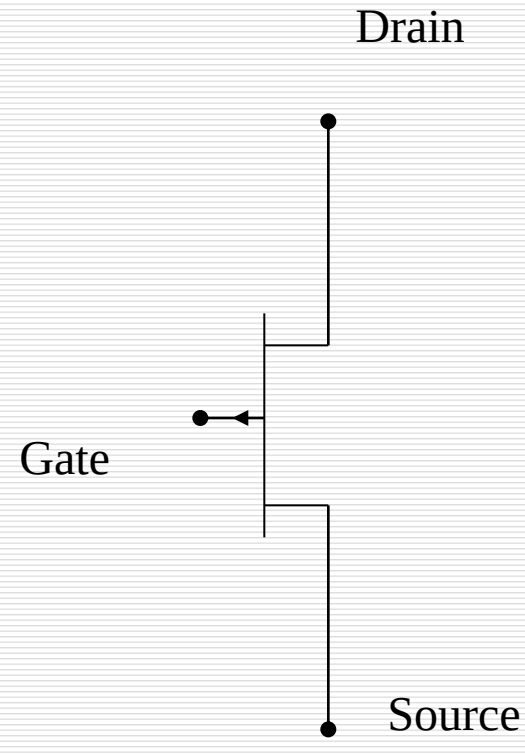
- There are 2 types of JFET
 - n-channel JFET
 - p-channel JFET

 - Three Terminal
 - Drain – D
 - Gate -G
 - Source – S
-

SYMBOLS



n-channel JFET



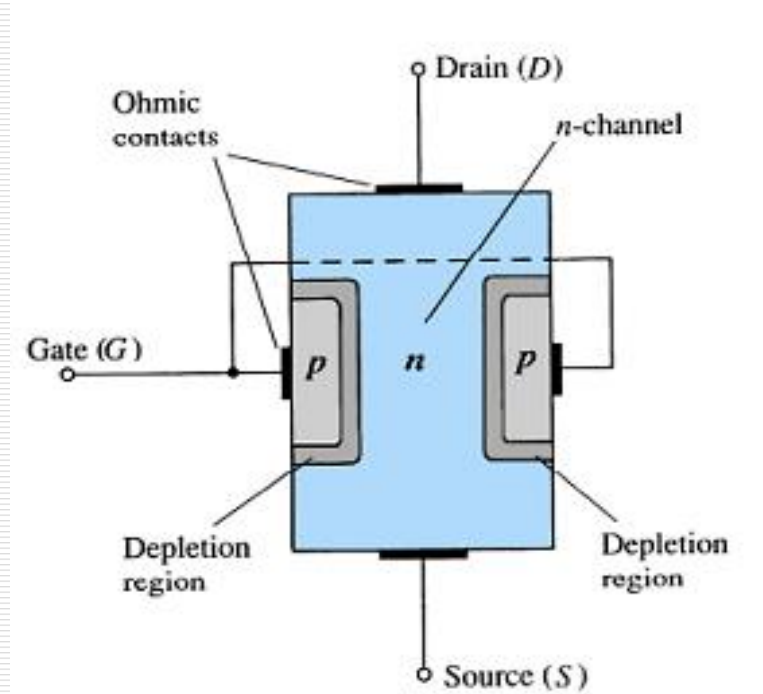
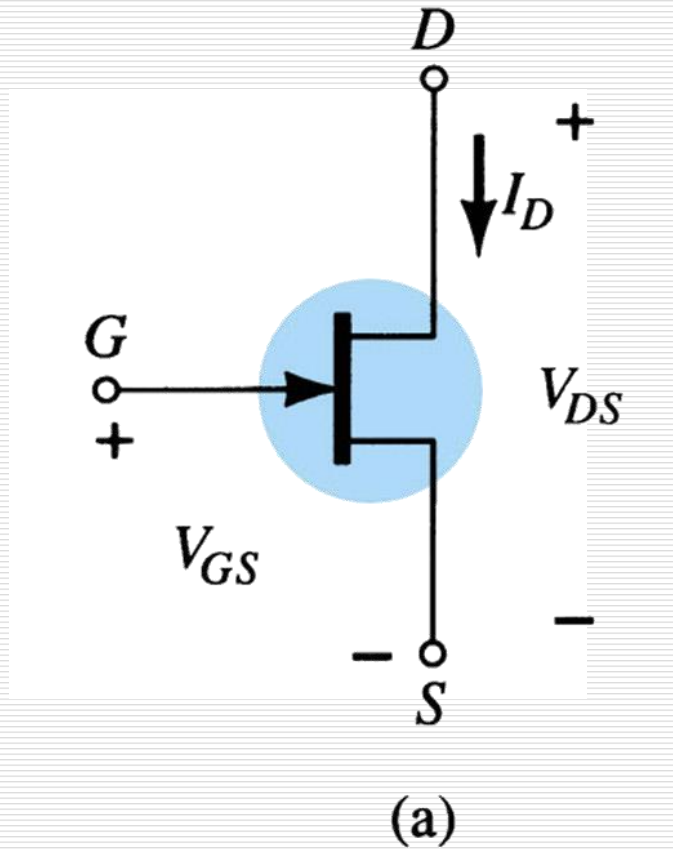
p-channel JFET

N-channel JFET

□ N channel JFET:

- Major structure is n-type material (channel) between embedded p-type material to form 2 p-n junction.
 - In the normal operation of an n-channel device, the Drain (D) is positive with respect to the Source (S). Current flows into the Drain (D), through the channel, and out of the Source (S)
 - Because the resistance of the channel depends on the gate-to-source voltage (V_{GS}), the drain current (I_D) is controlled by that voltage
-

N-channel JFET..

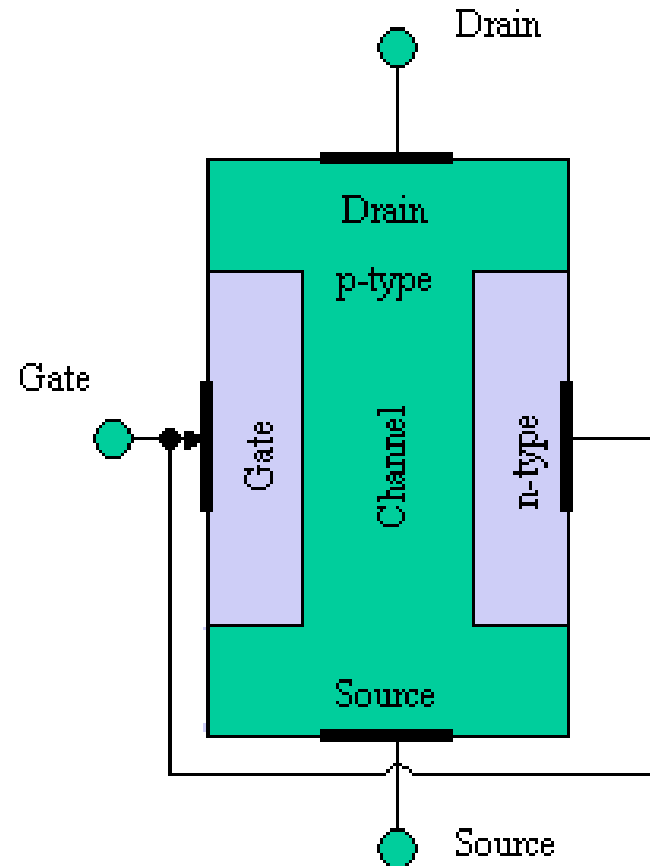
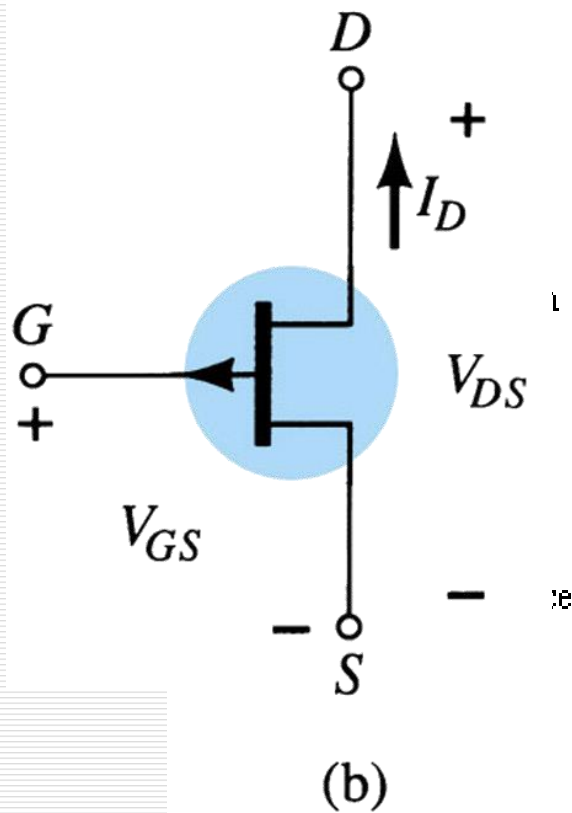


P-channel JFET

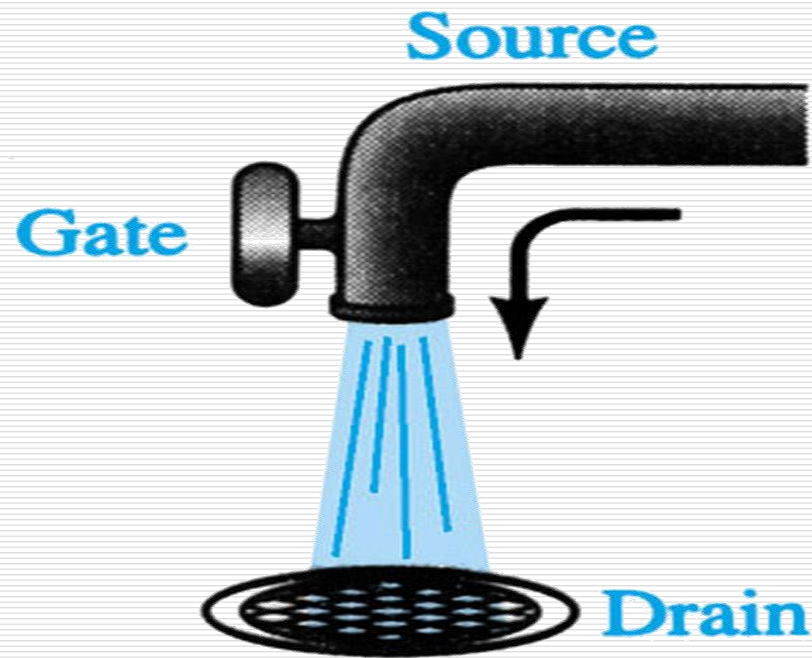
□ P channel JFET:

- Major structure is p-type material (channel) between embedded n-type material to form 2 p-n junction.
 - Current flow : from Source (S) to Drain (D)
 - Holes injected to Source (S) through p-type channel and flowed to Drain (D)
-

P-channel JFET..



Water analogy for the JFET control mechanism



JFET Operating Characteristics

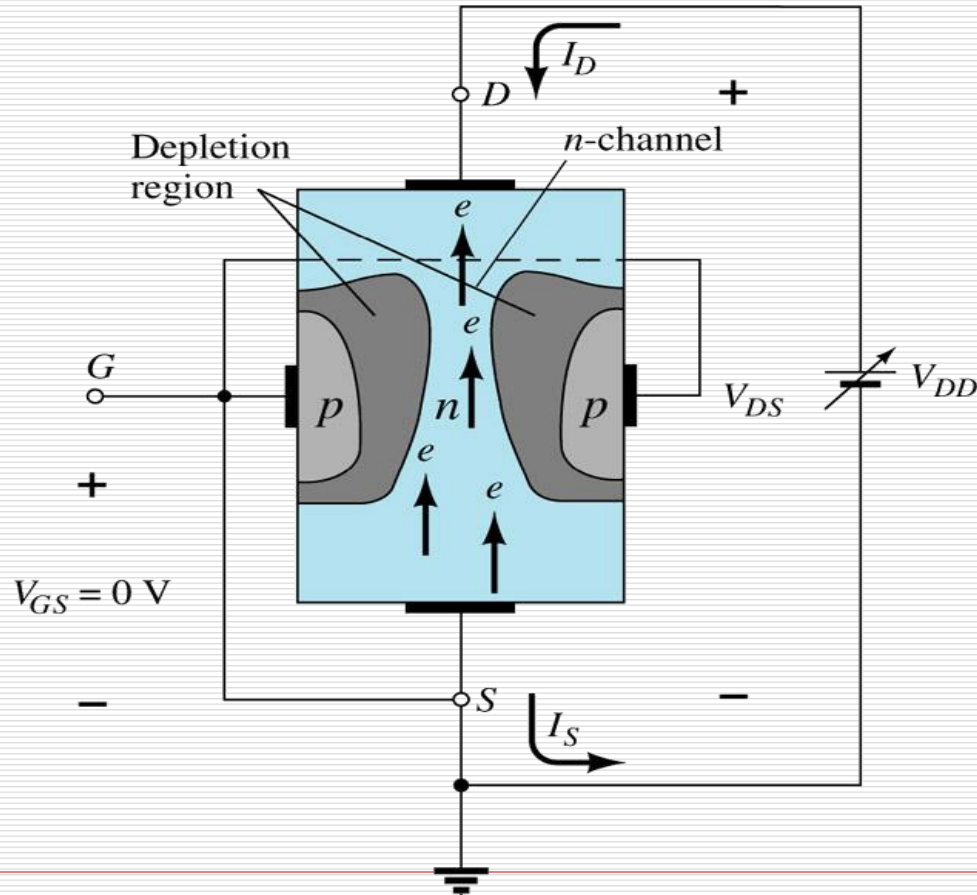
There are three basic operating conditions for a JFET:

- $V_{GS} = 0$, V_{DS} increasing to some positive value
- $V_{GS} < 0$, V_{DS} at some positive value
- Voltage-controlled resistor

JFET Characteristic for $V_{GS} = 0$ V and $0 < V_{DS} < |V_p|$

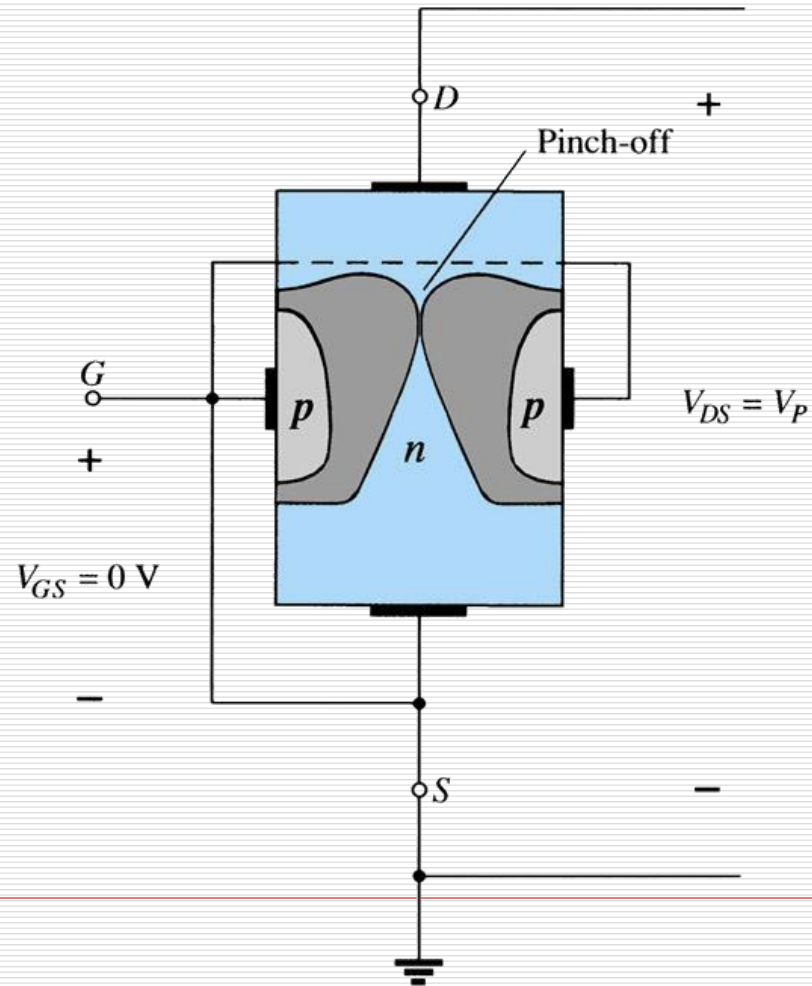
- To start, suppose $V_{GS}=0$
 - Then, when V_{DS} is increased, I_D increases.
Therefore, I_D is proportional to V_{DS} for small values of V_{DS}
 - For larger value of V_{DS} , as V_{DS} increases, the depletion layer become wider, causing the resistance of channel increases.
 - After the pinch-off voltage (V_p) is reached, the I_D becomes nearly constant (called as I_D maximum, I_{DSS} -Drain to Source current with Gate Shorted)
-

JFET for $V_{GS} = 0 \text{ V}$ and $0 < V_{DS} < |V_p|$



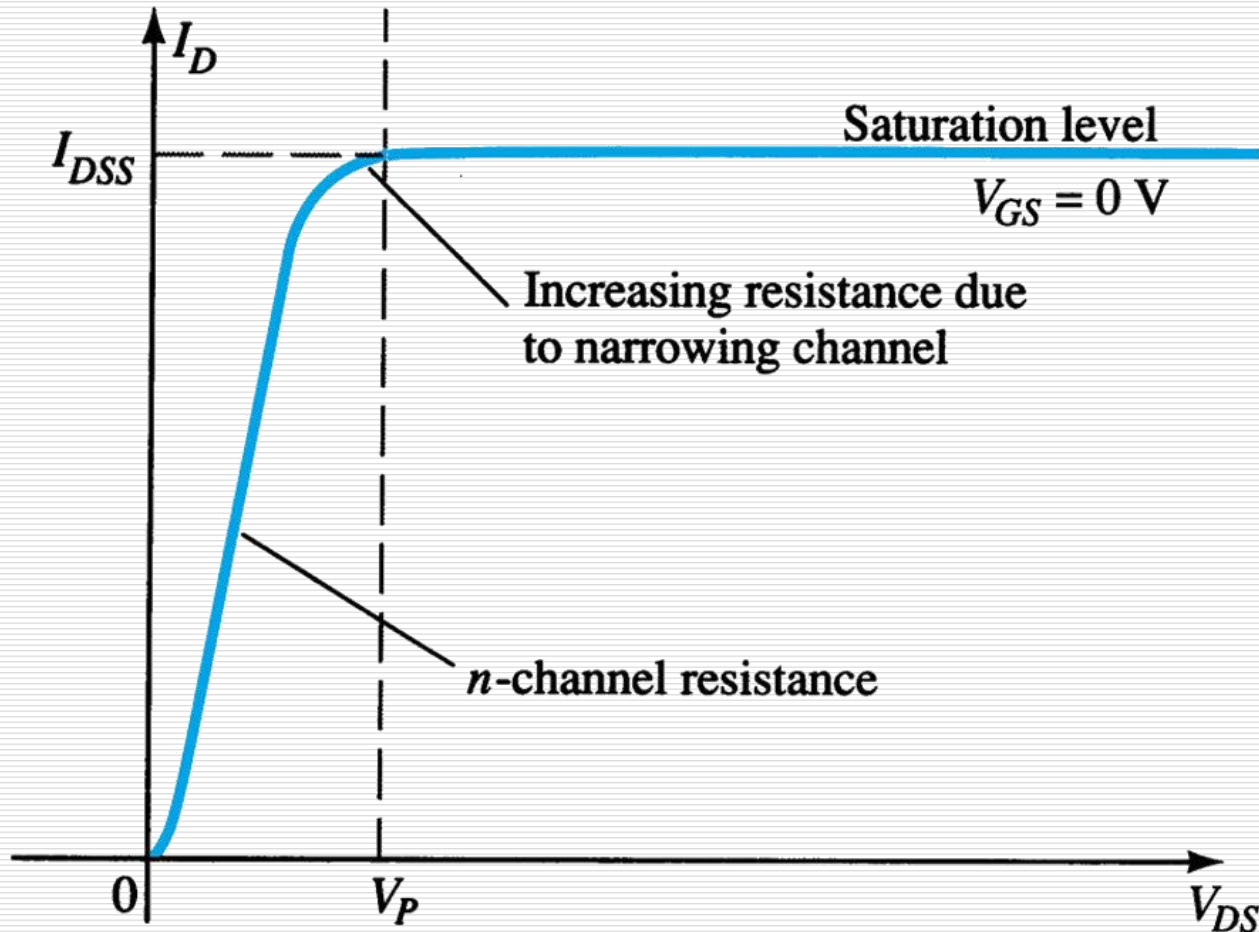
Channel becomes narrower as V_{DS} is increased

Pinch-off ($V_{GS} = 0 \text{ V}$, $V_{DS} = V_P$).



I_D versus V_{DS}

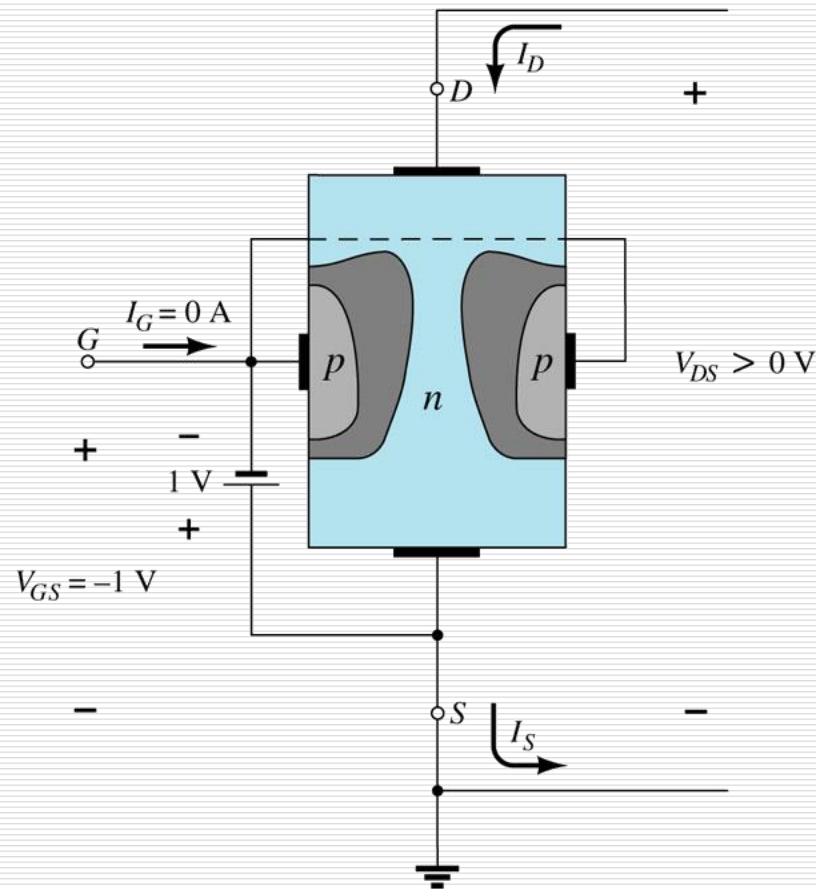
for $V_{GS} = 0$ V and $0 < V_{DS} < |V_p|$



JFET Characteristic Curve

JFET for $V_{GS} < 0$, V_{DS} at some positive value

(Application of a negative voltage to the gate of a JFET)



$V_{GS} < 0$, V_{DS} at some positive value

JFET Characteristic Curve..

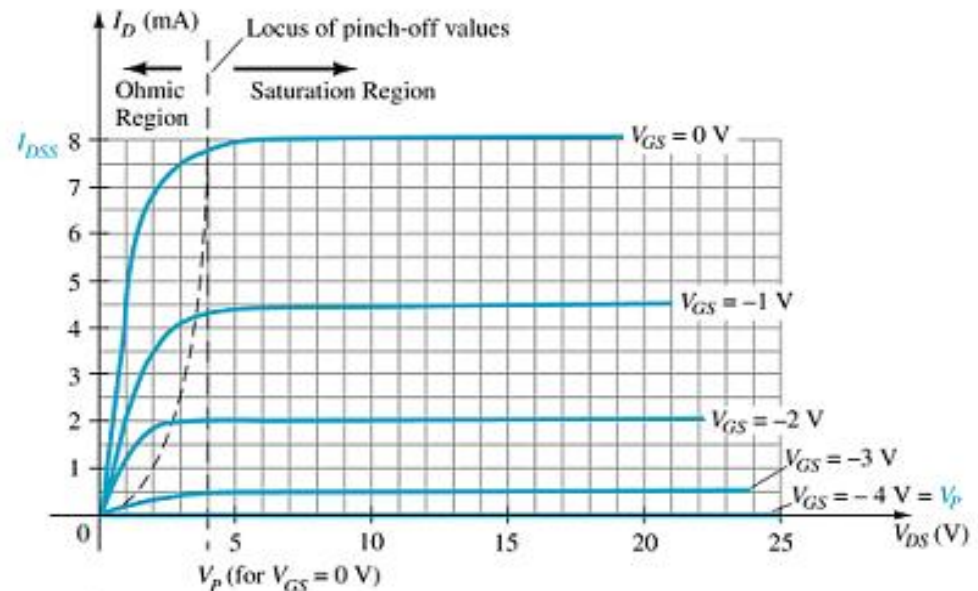
- ❑ For negative values of V_{GS} , the gate-to-channel junction is reverse biased even with $V_{DS}=0$
 - ❑ Thus, the initial channel resistance of channel is higher.
 - ❑ The resistance value is under the control of V_{GS}
 - ❑ If $V_{GS} = \text{pinch-off voltage}(V_P)$
The device is in **cutoff** ($V_{GS}=V_{GS(\text{off})} = V_P$)
 - ❑ The region where I_D constant – The **saturation/pinch-off region**
 - ❑ The region where I_D depends on V_{DS} is called the **linear/ohmic region**
-

$V_{GS} < 0$, V_{DS} at some positive value

JFET Operating Characteristics

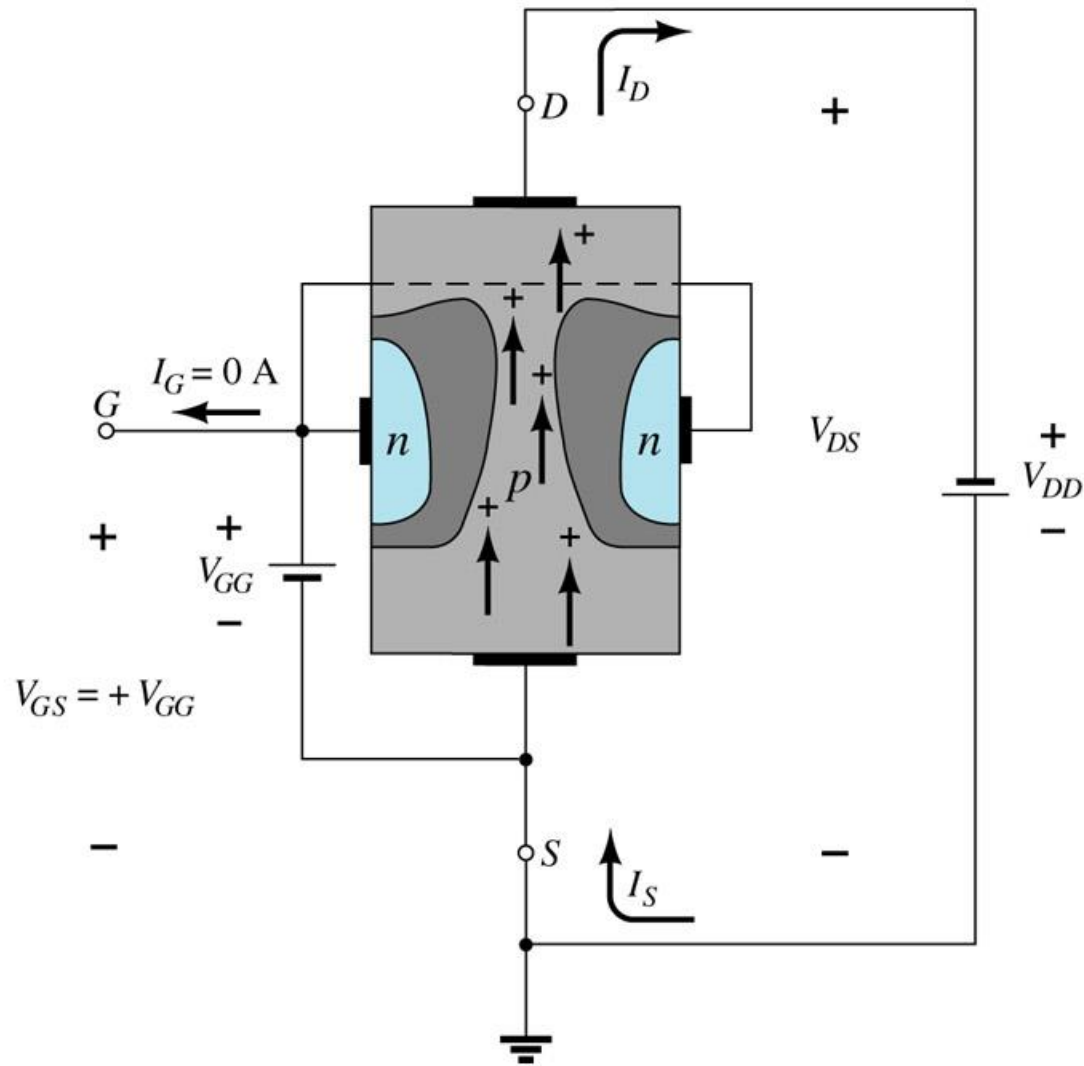
As V_{GS} becomes more negative:

- The JFET experiences pinch-off at a lower voltage (V_P).
- I_D decreases ($I_D < I_{DSS}$) even though V_{DS} is increased.
- Eventually I_D reaches 0 A. V_{GS} at this point is called V_P or $V_{GS(off)}$.

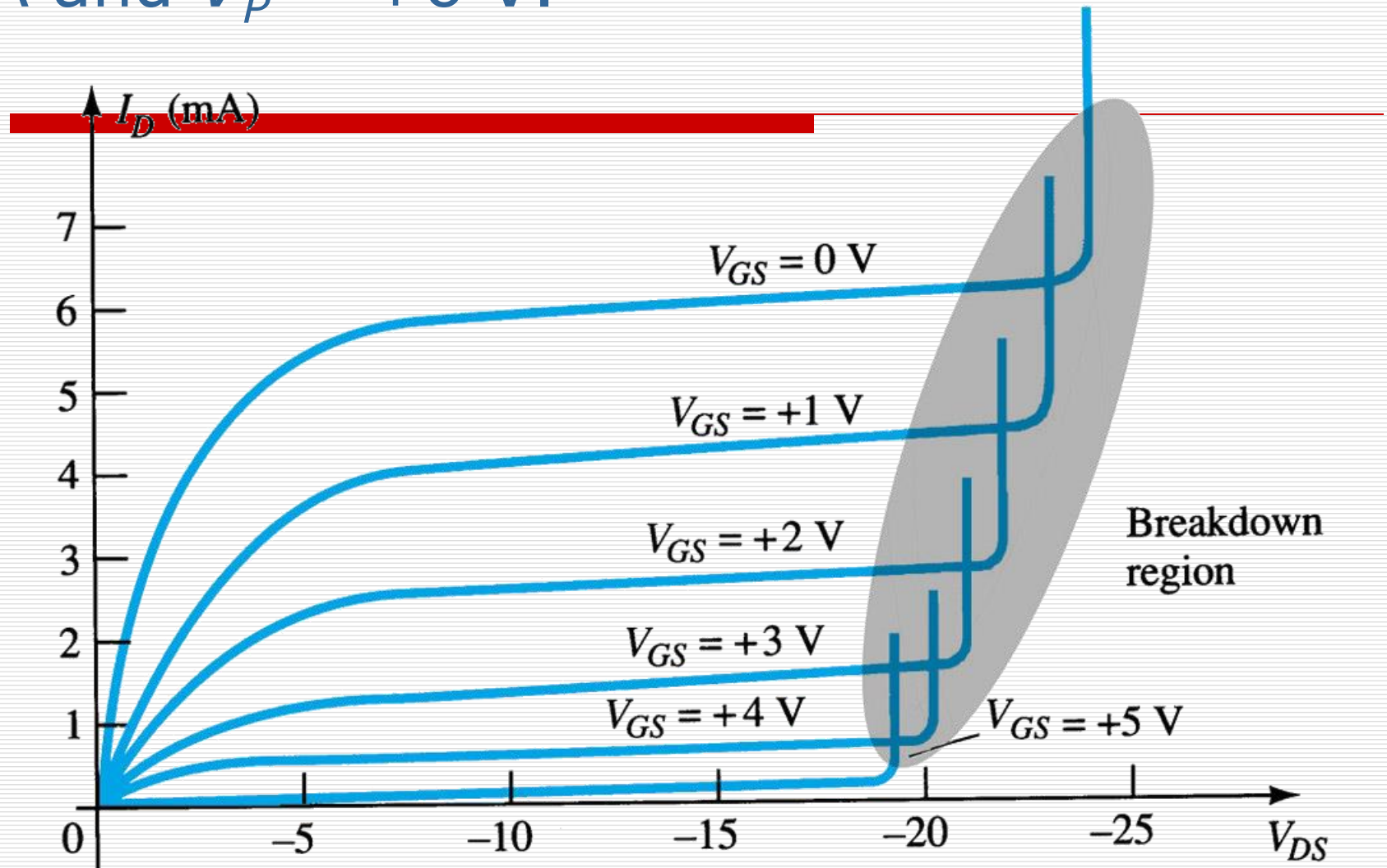


Also note that at high levels of V_{DS} the JFET reaches a breakdown situation. I_D increases uncontrollably if $V_{DS} > V_{DSmax}$.

p -Channel JFET

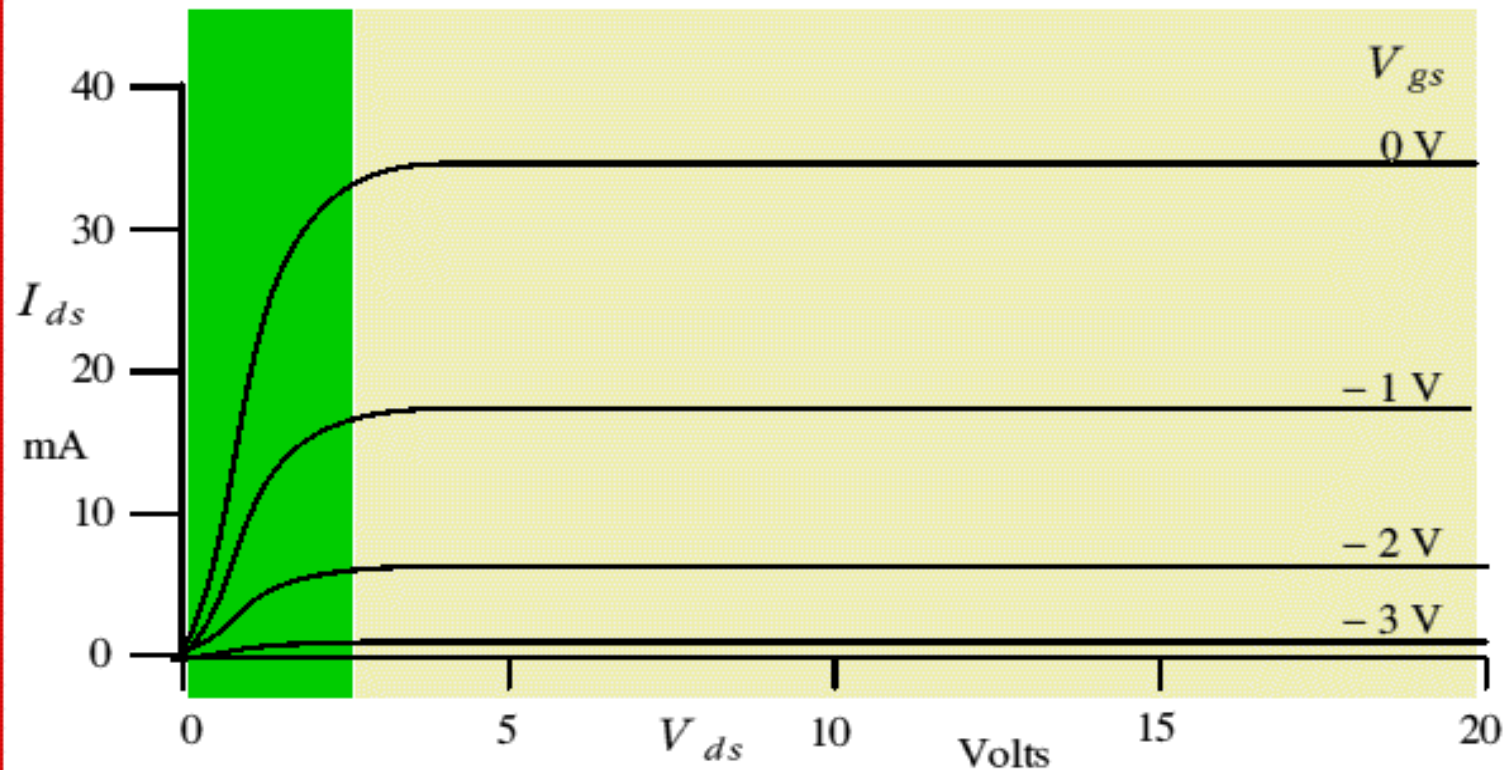


p -Channel JFET characteristics with $I_{DSS} = 6$ mA and $V_P = +6$ V.



Characteristics for n-channel JFET

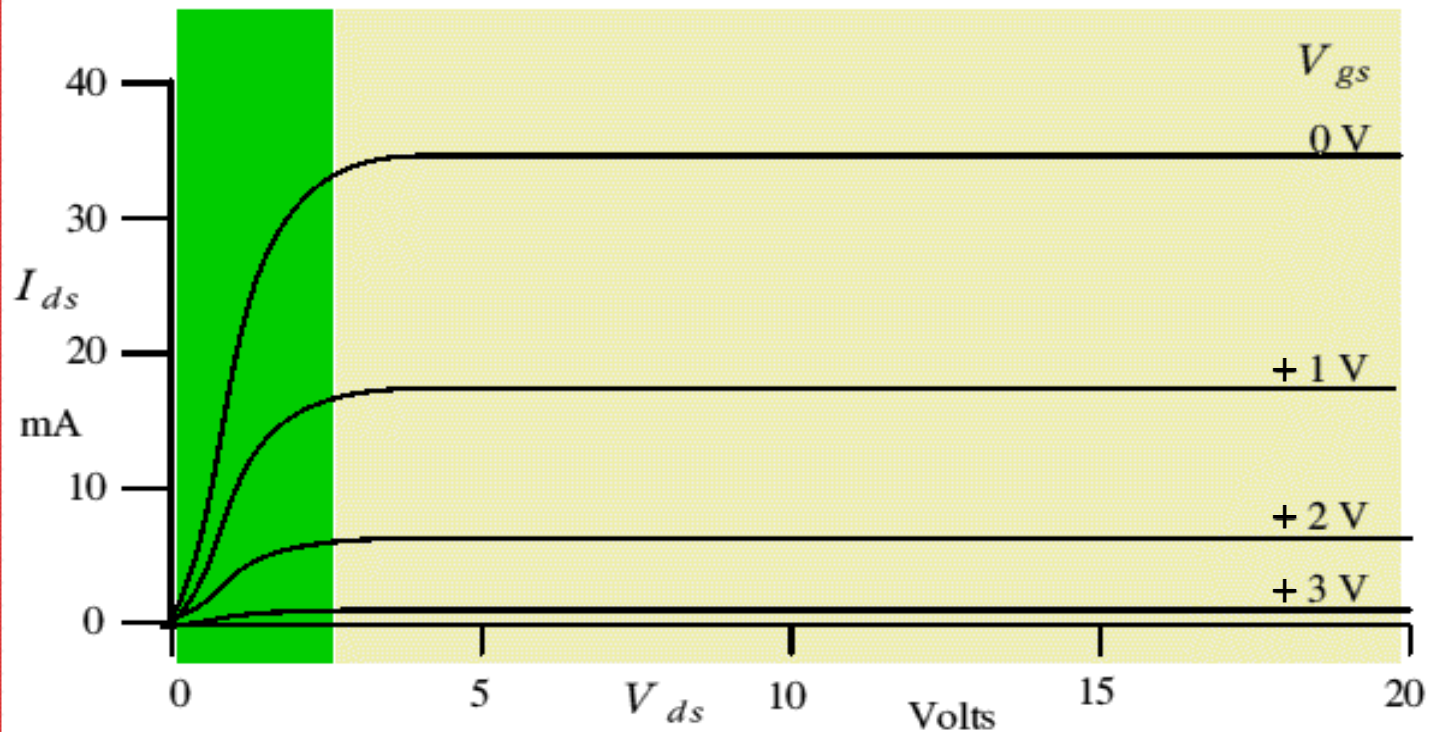
I_{ds} = drain-source current V_{ds} = drain-source voltage V_{gs} = gate-source voltage



'Characteristic curves for a typical N-channel JFET.

Characteristics for p-channel JFET

I_{ds} = drain-source current V_{ds} = drain-source voltage V_{gs} = gate-source voltage



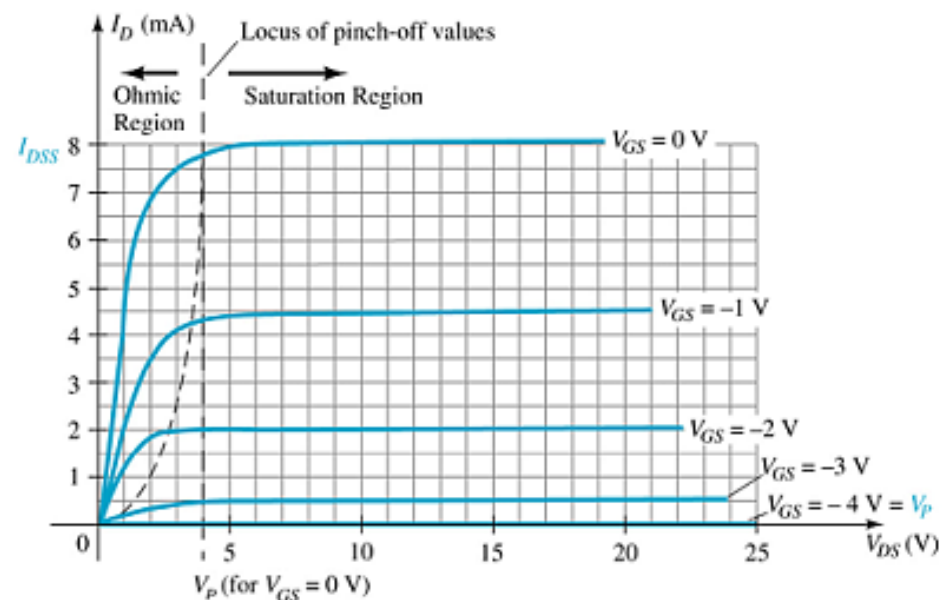
'Characteristic curves for a typical p-channel JFET.

JFET Operating Characteristics: Voltage-Controlled Resistor

The region to the left of the pinch-off point is called the **ohmic region**.

The JFET can be used as a variable resistor, where V_{GS} controls the drain-source resistance (r_d). As V_{GS} becomes more negative, the resistance (r_d) increases.

$$r_d = \frac{r_o}{\left(1 - \frac{V_{GS}}{V_P}\right)^2}$$



Transfer Characteristics

The input-output transfer characteristic of the JFET is not as straight forward as it is for the BJT. In BJT:

$$I_C = \beta I_B$$

which β is defined as the relationship between I_B (input current) and I_C (output current).

Transfer Characteristics..

In JFET, the relationship between V_{GS} (input voltage) and I_D (output current) is used to define the transfer characteristics. It is called as Shockley's Equation:

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2 \quad V_P = V_{GS (OFF)}$$

The relationship is more complicated (and not linear)

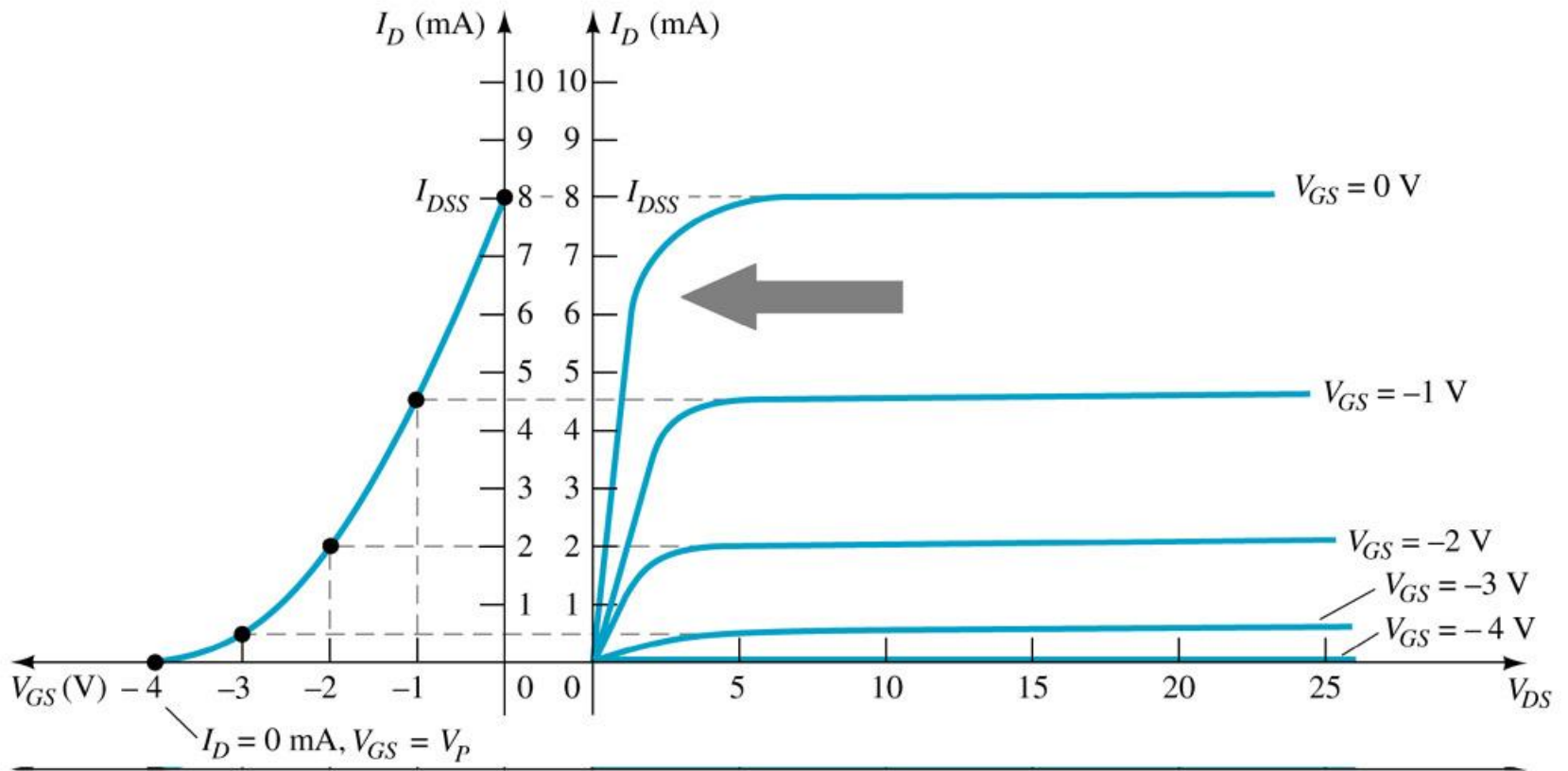
As a result, FET's are often referred to a square law devices

Transfer Characteristics...

- Defined by Shockley's equation:

$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_{GS(off)}} \right)^2 \qquad V_P = V_{GS(off)}$$

- Relationship between I_D and V_{GS} .
 - Obtaining transfer characteristic curve axis point from Shockley:
 - When $V_{GS} = 0 \text{ V}$, $I_D = I_{DSS}$
 - When $V_{GS} = V_{GS(off)}$ or V_P , $I_D = 0 \text{ mA}$
-



JFET Transfer Characteristic Curve

JFET Characteristic Curve

Exercise 1

Sketch the transfer defined by

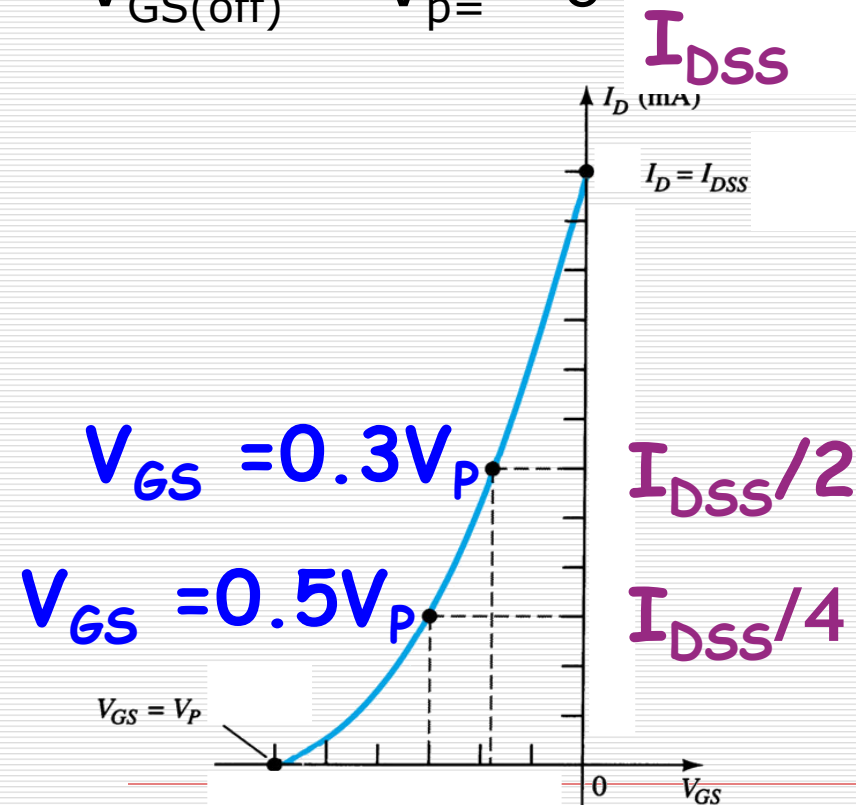
$$I_{DSS} = 12 \text{ mA dan } V_{GS(off)} = -6$$

V_{GS}	I_D
0	I_{DSS}
$0.3V_p$	$I_{DSS}/2$
$0.5V_p$	$I_{DSS}/4$
V_p	0 mA

$$V_{GS} = V_P \left(1 - \sqrt{\frac{I_D}{I_{DSS}}} \right)$$
$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2$$

Exercise 1

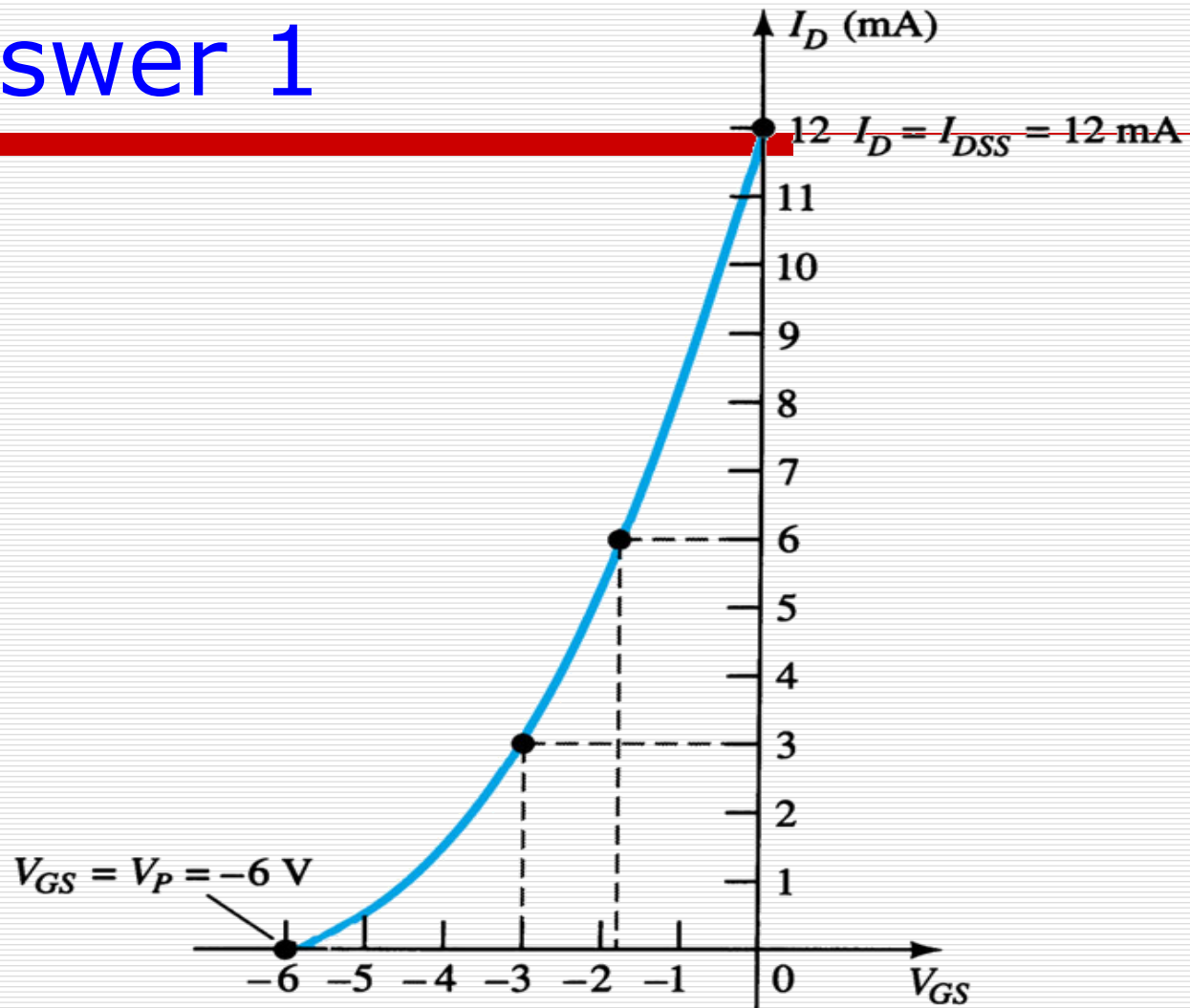
Sketch the transfer defined by $I_{DSS} = 12 \text{ mA}$ dan $V_{GS(off)} = V_p = -6$



$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2$$

$$V_{GS} = V_P \left(1 - \sqrt{\frac{I_D}{I_{DSS}}} \right)$$

Answer 1



Exercise 2

Sketch the transfer defined by

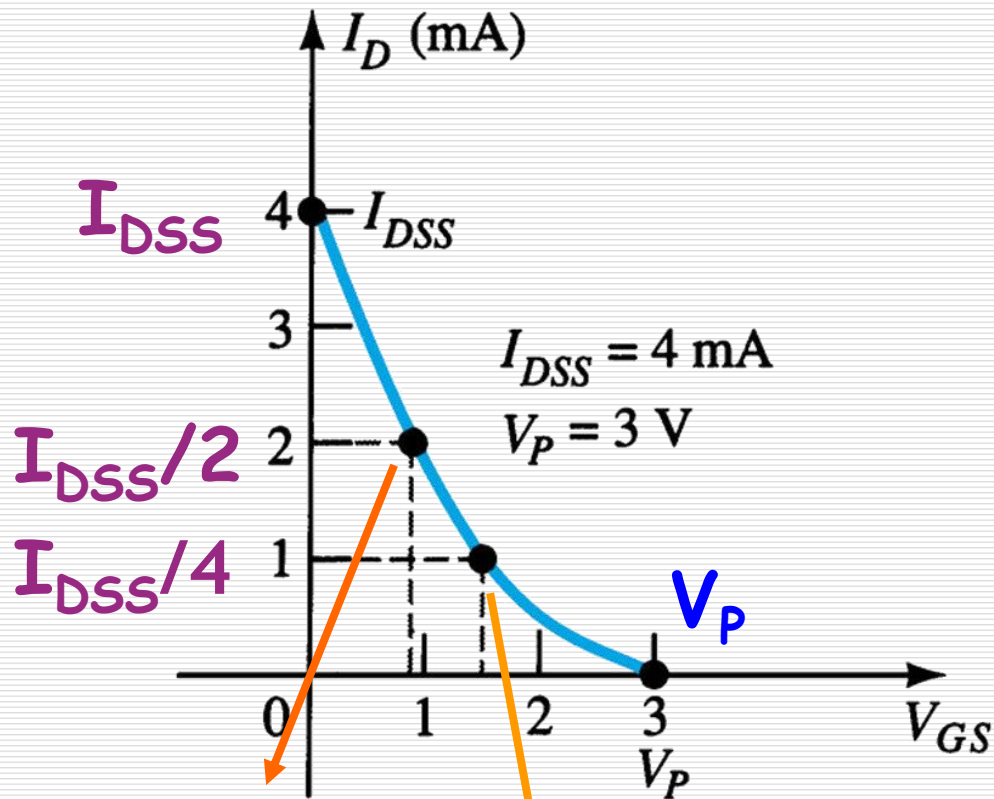
$$I_{DSS} = 4 \text{ mA dan } V_{GS(off)} = 3 \text{ V}$$

V_{GS}	I_D
0	I_{DSS}
$0.3V_p$	$I_{DSS}/2$
$0.5V_p$	$I_{DSS}/4$
V_p	0 mA

$$V_{GS} = V_P \left(1 - \sqrt{\frac{I_D}{I_{DSS}}} \right)$$
$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2$$

Exercise 2

Sketch the transfer defined by
 $I_{DSS} = 4 \text{ mA}$ and $V_{GS(off)} = 3 \text{ V}$



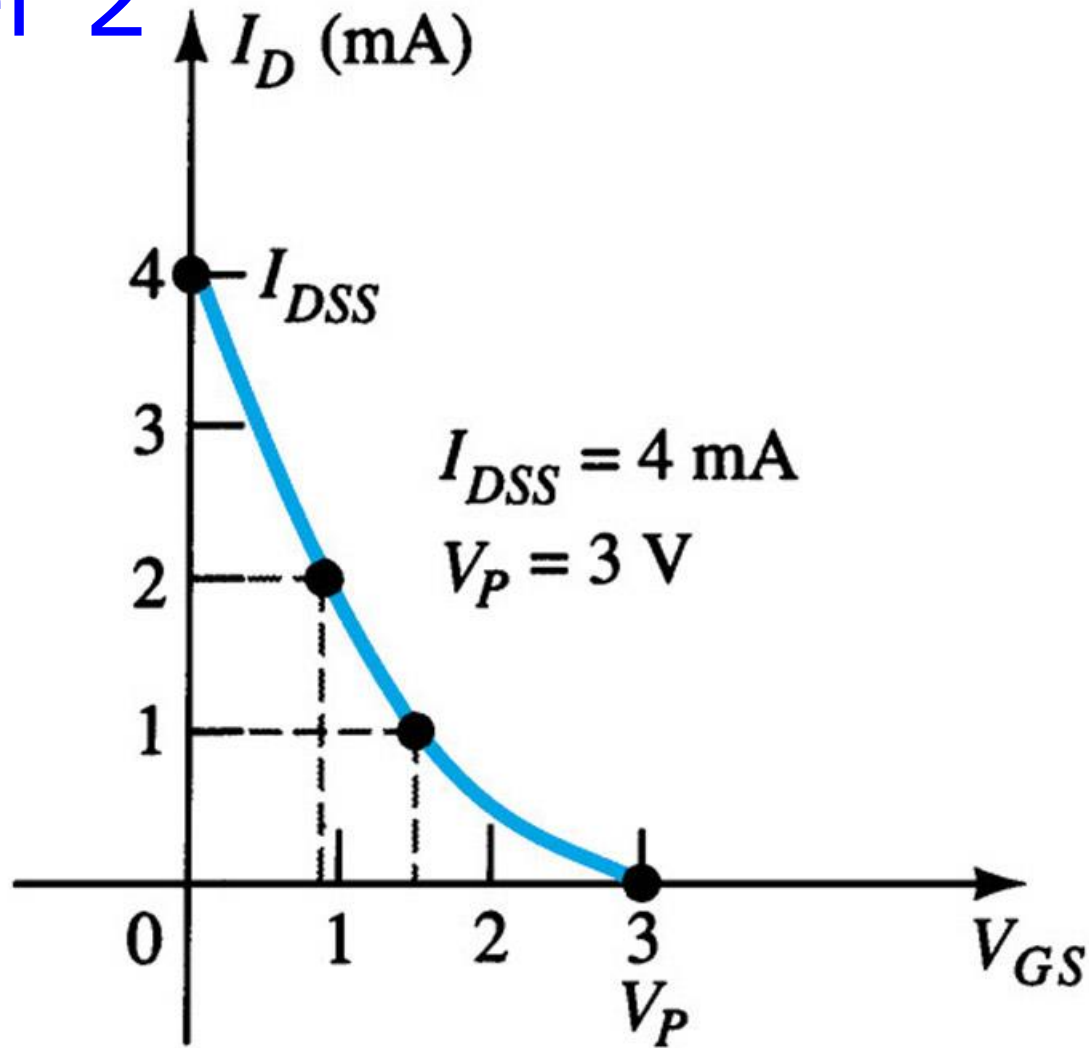
$$I_D = I_{DSS} \left(1 - \frac{V_{GS}}{V_P} \right)^2$$

$$V_{GS} = V_P \left(1 - \sqrt{\frac{I_D}{I_{DSS}}} \right)$$

$$V_{GS} = 0.3V_P$$

$$V_{GS} = 0.5V_P$$

Answer 2



A JFET has a drain current of 5 mA.

Exercise 3 $I_{DSS} = 10 \text{ mA}$ and $V_{GS(off)} = -6\text{V}$. Find the value of (i) V_{GS} and (ii) V_P

Solution.

$$I_D = I_{DSS} \left[1 - \frac{V_{GS}}{V_{GS(off)}} \right]^2$$

or

$$5 = 10 \left[1 + \frac{V_{GS}}{6} \right]^2$$

or

$$1 + \frac{V_{GS}}{6} = \sqrt{5/10} = 0.707$$

(i) $\therefore$

$$V_{GS} = -1.76 \text{ V}$$

(ii) and

$$V_P = -V_{GS(off)} = 6 \text{ V}$$

Exercise 4

Find V_{DS} and V_{GS} .

Given that $I_D = 5 \text{ mA}$

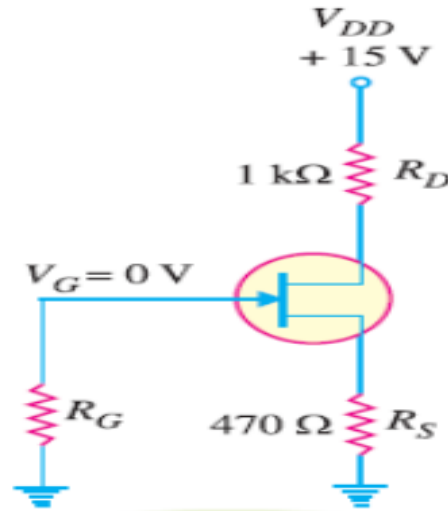


Fig. 5

Solution.

and

$$V_S = I_D R_S = (5 \text{ mA}) (470 \Omega) = 2.35 \text{ V}$$

$$\begin{aligned} V_D &= V_{DD} - I_D R_D \\ &= 15\text{V} - (5 \text{ mA}) \times (1 \text{ k}\Omega) = 10\text{V} \end{aligned}$$

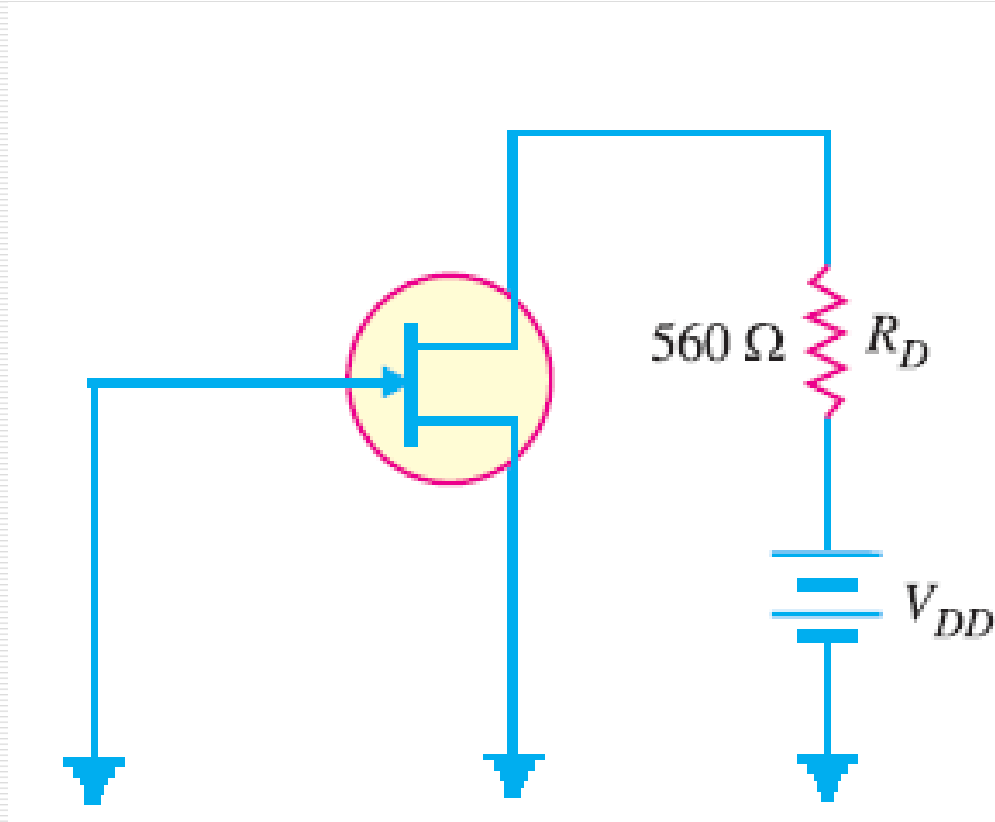
$$\therefore V_{DS} = V_D - V_S = 10\text{V} - 2.35 \text{ V} = \mathbf{7.65 \text{ V}}$$

Since there is no gate current, there will be no voltage drop across R_G and $V_G = 0$.

$$\text{Now } V_{GS} = V_G - V_S = 0 - 2.35\text{V} = \mathbf{-2.35 \text{ V}}$$

For the JFET in Fig., $V_{GS(off)} = -4V$ and $I_{DSS} = 12\text{ mA}$. Determine the minimum value of V_{DD} required to put the device in the constant-current region of operation.

Exercise 5



For the JFET in Fig., $V_{GS(off)} = -4V$ and $I_{DSS} = 12\text{ mA}$. Determine the minimum value of V_{DD} required to put the device in the constant-current region of operation.

Exercise 5

Solution. Since $V_{GS(off)} = -4V$, $V_P = 4V$. The minimum value of V_{DS} for the JFET to be in constant-current region is

$$V_{DS} = V_P = 4V$$

In the constant current region with $V_{GS} = 0V$,

$$I_D = I_{DSS} = 12\text{ mA}$$

Applying Kirchhoff's voltage law around the drain circuit, we have,

$$\begin{aligned} V_{DD} &= V_{DS} + V_{R_D} = V_{DS} + I_D R_D \\ &= 4V + (12\text{ mA})(560\Omega) = 4V + 6.72V = \mathbf{10.72V} \end{aligned}$$

This is the value of V_{DD} to make $V_{DS} = V_P$ and put the device in the constant-current region.